European Commission

Evaluation of the EU-India Science & Technology Agreement



Research and Innovation

Evaluation of the EU-India Science and Technology Agreement
Edited by the Global Service Facility (GSF) in Support of the Global Approach in Research and Innovation
GSF-11 TTC India" Support for the implementation of the EU-India Trade and Technology Council (TTC) Working Group 2 on green and clean energy technologies."

European Commission
Directorate-General for Research and Innovation
Unit DDG2.04 — International Cooperation -Asia, Pacific, Africa and the Middle East
Contact Antonio Marquez Camacho
Email *Antonio.Marquez-Camacho@ec.europa.eu*
RTD-PUBLICATIONS@ec.europa.eu
European Commission
B-1049 Brussels

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EUROPEAN COMMISSION

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Authors: Stephan Kreutzer, Theresa Witt, Tobias Kunde

2025

Directorate-General for Research and Innovation
EU-India Science and Technology Agreement

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1. Introduction and context of the review

1.1. The purpose of this review

This review evaluates the impact of the current **EU-India Science and Technology (S&T) Agreement**, set to expire on 17 May 2025. At the 14th meeting of the EU-India Joint Steering Committee on Science & Technology Cooperation held on 25 September 2024, both sides agreed to renew the Agreement for another period of five years without any amendments. The European Commission may now propose to the Council of the EU the renewal of the S&T agreement before its expiry date.

This review is **structured** as follows.

- Section 1 summarises the **purpose of this review** and the **context** in which it was conducted.
- Section 2 sketches out the **evolution of EU-India S&T collaboration** since the last review took place in 2020, focusing on Indian participation in the European research framework programmes and on human capital development.
- Section 3 dives deeper into **key topics of mutual interest** to the EU and India, and highlights cooperation results.
- Section 4 describes the **role of the Trade and Technology Council (TTC)** in underpinning EU-India S&T collaboration.
- Section 5 presents overarching **conclusions** from this review and **recommendations** for future cooperation.

1.2. Context of the review of the EU-India S&T Agreement

India is a **key partner of the EU**. Both have **common interests** in ensuring security, prosperity and sustainable development based on shared values, including a commitment to democracy, promoting human rights, and a rules-based global order. Strengthening this partnership is thus a priority for the EU. The **cooperation** between the EU and India focusses on key issues of common strategic importance, including trade, technology and security, particularly with regard to obstacles related to trade in technologies necessary to decarbonise economic production. In this context, **cooperation in S&T** is important to unlock the full potential of EU-India relations. The S&T cooperation between the EU and India unfolds within a **framework of several strategies and agreements**:

- The **EU-India Strategic Partnership**, renewed and endorsed at the EU-India Summit on 15 July 2020, provides a roadmap guiding joint cooperation activities until 2025, in particular in the areas of health research; bio-economy; digital technologies; green technologies with a focus on climate change and energy, resources efficiency and circular economy; polar sciences; social sciences and the humanities; mobility of scientists; gender equality; and strengthening the innovation dimension of the partnership through networking between startups. Research and innovation (R&I) is acknowledged as an important element of EU-India cooperation (points 72 to 84 of the document)[1]. The EU-India Strategic Partnership makes explicit reference to the S&T Agreement under review here (see point 72) and highlights the objective of deepening the cooperation through the EU R&I framework programmes ('Horizon Europe' and 'Horizon 2020') and a range of programmes and funding schemes run by various governmental departments in India on the basis of co-funding.

[1] EU (2020). EU-India Roadmap 2025

- The EU's approach to S&T cooperation with India is strategically underpinned by the **EU's R&I Strategy 2024**[2] and its **Global Approach to R&I**[3] which both foresee strengthening the EU partnerships and stepping up cooperation with India to jointly address global challenges and support India's sustainable modernisation process.
- S&T cooperation, especially concerning trade in technology, also needs to be seen in the context of ongoing negotiations between the EU and India to conclude a Free Trade Agreement (FTA) and an investment protection agreement (IPA).[4]
- On 8 May 2021, the **EU-India Connectivity Partnership** was launched in order to increase human capital development and people-to-people mobility of students and researchers in both directions. In addition, the European Commission has implementation agreements with the Indian Science and Engineering Research Board (SERB) and the Indian Council on Social Sciences Research (ICSSR) in place to provide short-term mobility opportunities for European Research Council (ERC) fellows. Human capital development was further strengthened when on 27 August 2024, a new scheme was announced to top up selected Marie Skłodowska-Curie Actions (MSCA) Staff Exchanges projects, enabling Indian entities to conduct joint research projects with European partners and second their staff to Europe-based organisations.[5]
- Lastly, in 2023 the **Trade and Technology Council** (**TTC)** with India was established, with one Working Group focusing on Green and Clean Energy Technologies and R&I cooperation (see chapter 4 for further information).

Overall, R&I cooperation is guided by the **Agreement on Scientific and Technical Cooperation**[6] (henceforth: the agreement), which is the subject of this review. The agreement was signed in 2001 and has been renewed several times. The last extension for another five years was approved on 17 May 2020. The **Joint Science and Technology Cooperation Committee (JSTCC)** is co-chaired by the Director General of Research and Innovation of the European Commission and the Secretary of the Ministry of Science and Technology of the Government of India. The JSTCC comes together generally on an 18-months interval basis to review progress and set future cooperation objectives. Meetings alternate between New Delhi and Brussels.

- On **12 February 2021**, the 13th **JSTCC meeting** took place online. Participants noted the success of flagship R&I calls on water, vaccines and smart grids. They also underlined the need for more balanced co-funding rules, i.e. by establishing an Implementation Arrangement complementing the S&T Agreement, allowing for better involvement of both sides in the entire co-funding process. New topics of mutual interest identified in the meeting included mitigation of plastic litter, building up the circular economy and electrical vehicles.
- The latest, 14th **JSTCC meeting** took place in Brussels on **25 September 2024**. Both sides:
  - Reaffirmed the strong 24-year India-EU partnership in Science and Technology (S&T) cooperation and set the stage for continued collaboration. **Both sides agreed to renew the EU-India S&T Agreement for 2025-2030** without any amendments.
  - Reviewed progress under the TTC, particularly on preparations for coordinated calls under the Horizon Europe Work Programme 2025, focusing on marine plastic litter, recycling of batteries, and waste-to-green hydrogen. Prospects for future cooperation were also discussed, especially in the areas of wastewater technologies and E-mobility.

[2] European Commission (2020). Research and innovation strategy 2020-2024

[3] European Commission (n. d.). EU Global Approach to R&I

[4] European Commission (n. d.). Research and Innovation: EU-India

[5] European Commission (2024). Boost for Indian-EU research partnerships under new co-funding scheme within MSCA Staff Exchanges

[6] EU (2002). Agreement for scientific and technological cooperation between the European Community and the Government of the Republic of India

- Highlighted the importance of fostering multilateral dialogue, promoting gender equality, and integrating open science principles.
- Agreed to strengthen cooperation in emerging areas of mutual interest, such as health (personalised medicine), bioeconomy, biotechnology, food security, polar research, and information communication technology (ICT).
- Reviewed India's participation in Horizon 2020 and Horizon Europe which showcased 88 collaborative projects with a combined funding of approximately €452m. Key joint initiatives in water, vaccines, and smart grids were identified as flagship efforts, and expanding opportunities for human capital development, including through the MSCA-CSIR co-funding arrangement, was stressed.

Finally, it is important for this review to consider the conclusions and recommendations included in a previous review that took place prior to the **last renewal of the Agreement** in 2020.[7] That review noted that the partnership aims to address global societal challenges such as climate change, energy, water stress, health, and food security. The main instruments of cooperation cited in the document are the EU's R&I framework programmes, notably Horizon 2020 and its successor, Horizon Europe. The previous review also noted that funding modalities had shifted since 2014 from automatic EU funding to a co-funding mechanism (CFM). The review **concluded** that the Agreement had been pivotal in advancing EU-India R&I cooperation, addressing mutual interests, and creating a structured dialogue on participation and funding. India's strong performance in European Research Council (ERC) calls and the MSCA highlights its scientific excellence. The review **recommended** to further simplify and streamline funding modalities, avoid fragmentation and engage in strategic areas of cooperation, and to build upon momentum created by innovation pilot projects. It was also suggested to move beyond water, health and energy into areas such as ICT, circular economy, biodiversity, bio-economy, and cultural heritage, and to enhance two-way mobility (i.e. attracting more European researchers and technical experts to India).[8] The current review includes an assessment on whether and to what extent previous recommendations have been implemented.

In the following sections, this **review** takes stock of the progress of EU-India S&T cooperation over the past 4 years, focusing primarily on the Indian participation in Horizon 2020 and Horizon Europe and human capital development, joint activities in specific priority areas, as well as the work done in the TTC.

[7] European Commission (2020). Renewal of the Agreement for Scientific and Technological Cooperation between the EU and India

[8] Presentation made to the 13th EU-India Steering Committee meeting on Science and Technology Cooperation, 12 February 2021

2. Evolution of EU-India S&T cooperation

Throughout the period in scope of this review, India's collaboration with the EU on R&I took place through the **Horizon 2020** (2014-2020) and **Horizon Europe** (2021-27) framework programmes. R&I in both Horizon 2020 and Horizon Europe are promoted through **three main pillars**. Under Horizon 2020, these pillars were defined as *Excellent Science*, *Industrial Leadership*, and *Societal Challenges*. In Horizon Europe, the pillars were adapted to *Excellent Science*, *Global Challenges and Industrial Competitiveness*, and *Innovative Europe*. In the first pillar (Excellent Science), Indian and EU scientists and innovators benefit from collaboration through participation in capacity-building, human capital development and mobility actions facilitated by the ERC and MSCA. In the other two pillars, the participation of Indian individuals and entities is linked to fostering R&I initiatives and collaborative projects within their respective fields.

In the following, we describe the **key trends regarding EU-India S&T cooperation** in the review period, focussing on Horizon projects, and ERC and MSCA participation.

2.1. Cooperation under Horizon 2020 and Horizon Europe

Horizon 2020 and **Horizon Europe** are and were based on calls for proposals, most of which are open for collaboration with Indian partners. Indian entities can participate in projects in three ways: *joint or coordinated calls*, *earmarked calls* co-funded by the Indian Government, and *open calls*. Nevertheless, Indian participants are not automatically eligible for funding under Horizon Europe. Unless the call specifically indicates an exceptional EU funding, Indian participants must determine their own funding sources, which may include own financing, or funds from Indian ministries, agencies, foundations, or other organisations that support international R&I activities. The Department of Science and Technology (DST), the Department of Biotechnology (DBT) and the Ministry of Earth Sciences (MoES), Government of India, have set up a Co-Funding Mechanism (CFM) on the modalities of funding the successful Indian entities in selected Horizon 2020 projects.[9] This has continued under Horizon Europe, including in the latest Work Programme (WP) 2023-24.[10]

The following sub-sections provide figures showcasing **trends and patterns regarding India's participation in the EU's R&I programmes**.

[9] European Commission (2019). India-EU Co-Funding Announcement for Proposals under 'Horizon 2020' Work Programme

[10] European Commission (2023). Announcement on India-EU Cooperation on R&I under 'Horizon Europe' Framework Programme

2.2. Statistics on Indian participation in Horizon 2020 and Horizon Europe

Table 1 provides an overview of **Indian participation in Horizon 2020 and Horizon Europe from 2018-24**.[11]

	Total	Horizon 2020	Horizon Europe
1. OVERVIEW			
No. of projects with Indian participation	88	52	36
No. of Indian participations	204	161	43
Total EU contribution to projects (with Indian participation)	410.425.948,68 €	251.919.830,55 €	158.506.118,13 €
Total EU contribution to Indian participants	4.810.278,68 €	3.040.980,69 €	1.769.297,99 €
Total non-EU budget of Indian participants*	40.700.849,00 €	39.085.651,00 €	1.615.198,00 €
2. COLLABORATIVE ACTIONS (all except ERC & MSCA)			
No. of projects with Indian participation	57	36	21
No. of Indians participations in collaborative actions	169	143	26
Total EU contribution to projects (with Indian participation)	357.409.164,88 €	226.360.240,59 €	131.048.924,29 €
Total EU contribution to Indian participants	2.795.936,58 €	1.724.138,59 €	1.071.797,99 €
Total non-EU budget of Indian participants*	41.700.849,00 €	39.085.651,00 €	1.615.198,00 €
3. CO-FUNDED CALLS			
No. of projects with Indian participation*	8	5	3
No. of Indian participations in co-fund Projects*	14	7	7
Total EU contribution to projects (with Indian participation)	76.037.625,43 €	62.074.365,43 €	13.963.260,00 €
Total EU contribution to Indian participants	0	0	0
Total non-EU budget of Indian participants*	4.370.565,60 €	3.070.565,60 €	1.300.000,00 €
4. JOINT CALLS (Water, Vaccine, Small Grid)			
No. of projects with Indian participation*	12	12	0
No. of Indian participations in joint projects	107	107	0
Total EU contribution to projects (with Indian participation)	54.711.624,63 €	54.711.624,63 €	0,00 €
Total EU contribution to Indian participants	0,00 €	0,00 €	0,00 €
Total non-EU budget of Indian participants*	34.443.717,40 €	34.443.717,40 €	0,00 €
5. Marie Sklodowska-Curie Actions (MSCA)			
No. of projects with Indian participation (as partners)	27	11	17
Total number of individual Indian researchers (2022-2024)*	963	492	471
6. ERC			
No. of Indian participations (as partners)	8	7	1
No. of ERC grants (2018-2023)	46		
No. of main grants (Starting, Consolidator, Synergy)	36		
No. of Proof of Concept grants	10		

Participations: institutes that participate as international partner and as benificiary count as 2

*Total non-EU budget = Total Cost - Eu Contribution (lower bound value since the co-funded amount of most Indian partners is not specified when participating with own funding). Values for 2023 and discrepancies with data from EU R&I Projects Dashboard: data from informal communication with Government of India (GoI)

Table 1. Overview of Indian participations (2018-2024)[12]

11 The review considers data from 2018 onwards because this concerns projects that were still operating in 2020 when the time period within scope of this review began.

12 Data Source: EU R&I Projects. Dashboard. Accessed 15 of August 2024

The first section of Table 1 (Overview) shows that between 2018 and 2024, Indian entities participated in a total of 88 projects, with 52 under Horizon 2020 and 36 under Horizon Europe. In total, 204 Indian participants were involved, including 161 in Horizon 2020 and 43 in Horizon Europe. Most of these projects (169 out of 204 or 83%) were within collaborative actions (section 2 in Table 1), including projects with Indian entities as beneficiaries, third party and or international/associate partner status, such as in case of co-funded and joint calls. Most of those collaborative actions can be attributed to the pillar Societal Challenges in Horizon 2020, and to the pillar Global Challenges and European Industrial Competitiveness of Horizon Europe. The remaining 40% (35 out of 88) of projects in Horizon programmes (sections 5 and 6 in Table 1) sat in the first pillar of both Horizon programmes, Excellent Science, with 28 participations from Indian partners in collaborative actions, 471 Indian researchers involved in MSCA (between 2022-2024), and 8 participations through the ERC (between 2018-2024).

Looking at funding patterns, the table displays that under the Horizon programmes, **India played a significant role in these collaborations** as a third country. In all collaborative actions (except ERC & MSCA) since 2018, India contributed through co-funding with more than 12% (€41.7m out of €357.4m) in the total project funding. During Horizon 2020, this co-investment reached over 17% (€39.1m out of €226.4m) (section 2 of Table 1). The largest co-funding amount came from the joint calls on water, vaccine, and smart grid with a total Indian Budget of €34.4m (section 4 of Table 1). Nevertheless, this value decreased to a minimum of 1% (€1.6m out of €131m) under Horizon Europe (section 2 of Table 1). It is important to note that this proportion could potentially underestimate the Indian co-funding, since the amount requested by most Indian partners is not specified in the data available, and grants for most Horizon Europe calls have not been signed yet.

These developments in funding align with the **downward trend in the overall project budget** allocated to projects with Indian participants (see Total contribution EU to projects in Figure 1). A significant **decrease in the Indian budget** is observed from Horizon 2020 to Horizon Europe, with the higher Indian budget before 2022 primarily attributed to **joint calls**, except for 2020, when the complete absence of a joint call resulted in the lowest non-EU budget within Horizon 2020.

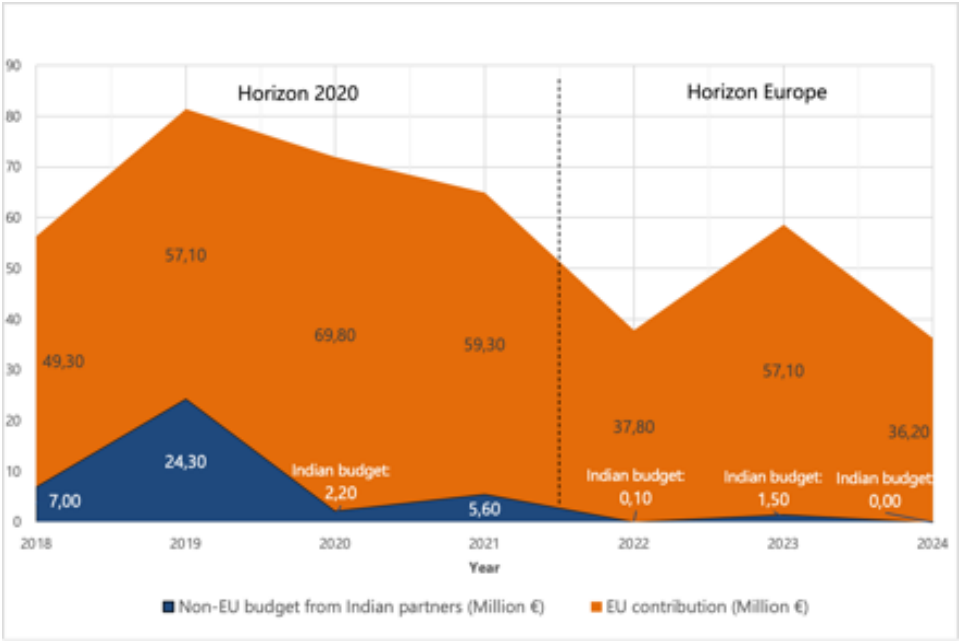


Figure 1. Trends in the Ratio of EU Contributions to Non-EU Indian Budget in R&I Projects[13]

[13] Low value in 2024, since Grants for co-fund calls have not been signed yet for 2024.

As noted, in Horizon Europe, co-funding of calls was prioritised, and the use of joint calls was discontinued. Following this, the financial participation of Indian entities, as well as the number of Indian participants in projects, declined compared to Horizon 2020 (see Figure 2).

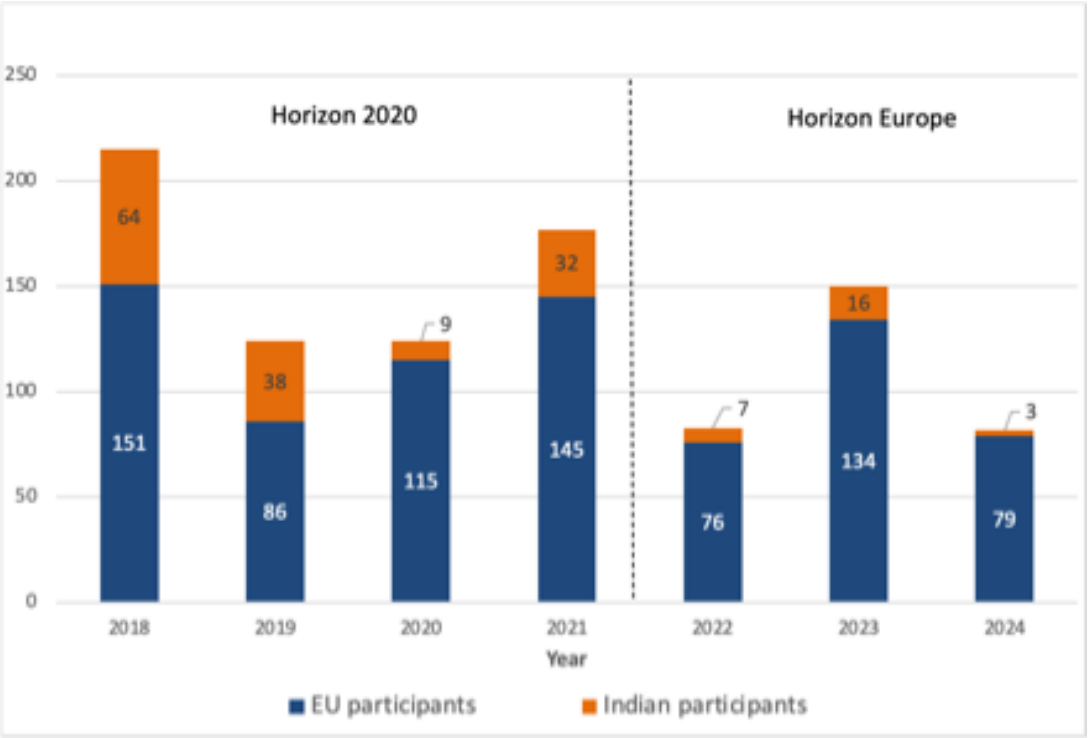
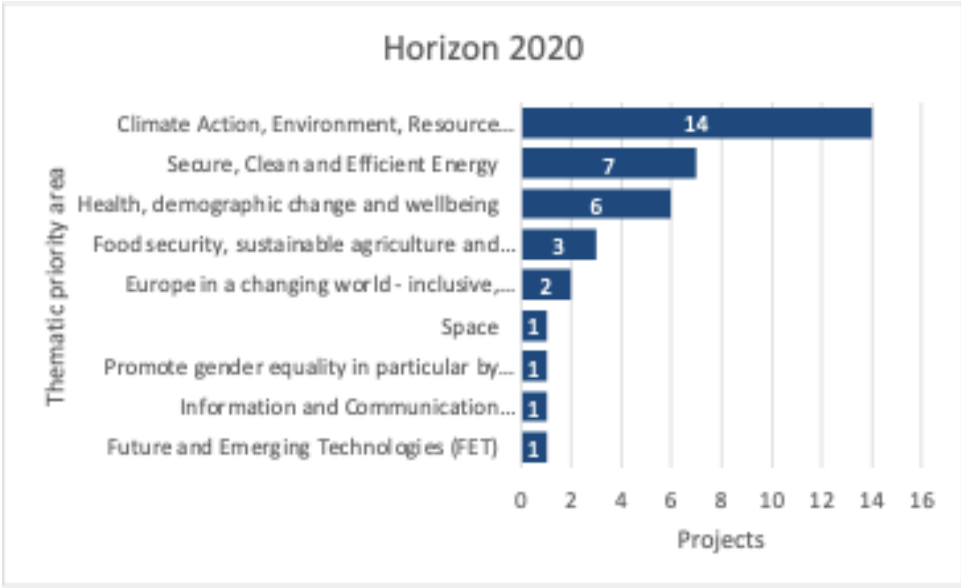


Figure 2. Trends in the Ratio of EU to Indian Participations in R&I Projects

Although there was a shift in the distribution of the thematic priority areas of R&I projects, the **primary focus areas remained largely consistent** over time (see Figure 3). Notably, environmental projects (encompassing Climate, Environment, Resources, and Raw Materials) emerged as the leading thematic priority under Horizon 2020, with more than twice the number of projects compared to the second priority area, Secure and Clean Energy. Even though the thematic area related to energy (Secure, Clean and Efficient Energy) was integrated with Climate, Energy and Mobility in Horizon Europe, a shift was evident with the majority of projects concentrating on Health. However, while Health was the primary focus, Climate, Energy, and Mobility remained important, representing the second most prioritised area. Moreover, despite the lower number of projects, the more even distribution of projects across thematic priority areas in Horizon Europe suggests a thematic diversification overall.



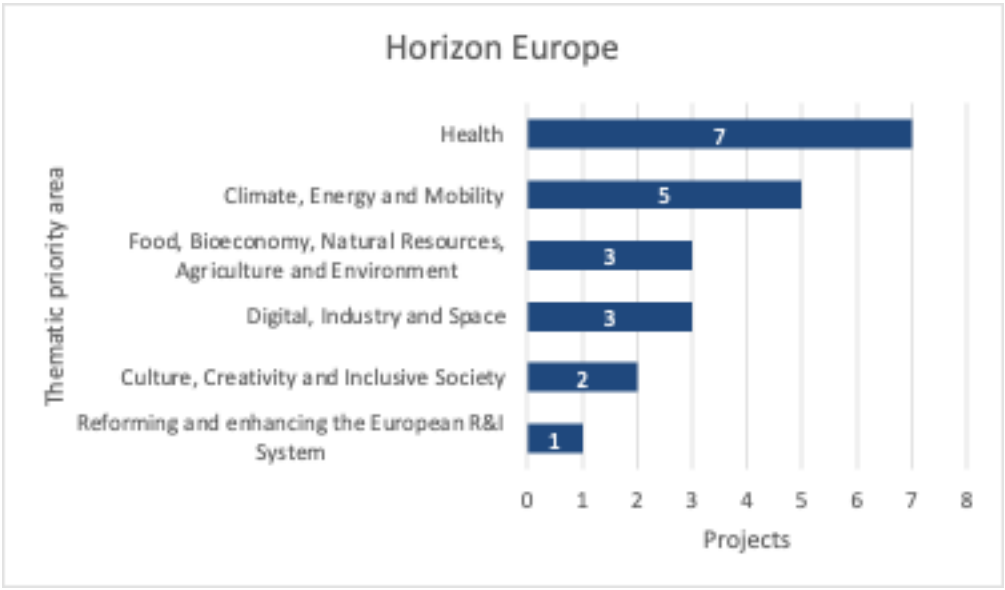


Figure 3. Thematic Priority Areas at Project Level (2018-2024)

2.3. Co-funded calls and Horizon Europe Work Programme 2023-2024

As part of the CFM, **Indian entities participated in calls for proposals** earmarked by Government of India's funding agencies, contributing to projects under both Horizon 2020 and Horizon Europe. From 2018 to 2021, under Horizon 2020, the Department of Biotechnology (DBT) and the Ministry of Earth Sciences (MoES) served as co-funding agencies. During this period, **five proposals with Indian participation** were selected: three in the areas of Climate Action, Environment, Resource Efficiency, and Raw Materials, and two in Health, Demographic Change, and Well-being. These projects were co-funded by Indian agencies with a total contribution of €3m.

Under the Horizon Europe Work Programme 2023-2024, **11 calls for proposals** were earmarked by India for co-funding, with the Department of Biotechnology (DBT), the Department of Science and Technology (DST), and the Ministry of Earth Sciences (MoES) serving as the funding agencies. At the time of this report's writing (November 2024), two calls were still under consideration, as the submission and selection phases had not yet concluded. For the remaining nine calls, Indian entities participated in 49 of the 256 proposals submitted. Out of these 49 proposals, seven with Indian participation were selected, resulting in a 14% success rate. These successful proposals focused on areas of artificial intelligence, battery technology, essential biogeochemical ocean variables, inland ice, and pandemic preparedness and response.

2.4. Lessons learnt from EU-India S&T cooperation

The EU-India R&I cooperation under Horizon 2020 and Horizon Europe was carried out through a co-funding mechanism, involving two approaches: (i) joint/flagship calls where Indian participation is mandatory, and (ii) open calls earmarked by the Government of India for co-funding. In both cases, selected projects were supported by the respective funding agencies of the European Commission and India. However, throughout the process of designing calls, project selection, and implementation, both sides encountered challenges. The following "Lessons Learned" are based on feedback from officials on both sides.:

- **Coordination and Communication:** Coordination between the European Commission (EC) and Indian co-funding organisations on call details and timelines as well as during evaluation and grant phases could be improved. This starts with internal coordination and alignment of departments on both sides. Early agreement on modalities and legal reviews is necessary to prevent delays in grant procedures.
- **Evaluation and Expert Panels:** The Indian side could be more closely involved in the selection of expert panels reviewing project applications for joint research. The rules for the selection of experts could be communicated more clearly. Evaluation summary reports should be shared by project coordinators with Indian participants in a timely manner so that Indian organisations may secure co-funding for selected projects.
- **Funding and Budgeting:** Delays in fund disbursement and project initiation by Indian partners slow down implementation.
- **Budget allocation:** In EU-India joint/coordinated calls, the EU provides a fixed funding amount per call and per project upfront, which is allocated to successful EU entities. However, the Government of India does not specify an absolute funding amount but instead indicates "matching funding," which often creates challenges.
- **Project Implementation and Review:** Joint reviews provide an opportunity to assess the impact and learn from completed projects. Evaluations during project implementation allow to change course if needed.
- **Participation and Visibility:** Indian partners are underrepresented in EU-managed calls partly due to Indian participation not being a mandatory requirement in the call text (except in the case of joint calls). In the case of most co-funded and earmarked calls in Horizon Europe, Indian participation is not mentioned explicitly as a targeted country, and this contributes to low participation rates for Indian entities. Conversely, when India is mentioned explicitly, as was the case in a recent call focused on Artificial Intelligence, European consortia end up including Indian entities in their proposals, therefore increasing the chances of them being part of selected projects. Therefore, more visibility of India as a targeted country in Horizon Europe calls may lead to increased overall participation of Indian entities in submitted proposals and in the end in selected projects.

2.5. Human Capital Development of researchers/scientists and innovators

The EU-India Connectivity Partnership[14] (adopted in 2021) includes a People-to-People segment which foresees advanced **cooperation on human capital development, capacity-building and mobility actions** for researchers, in particular through **MSCA, ERC** and **India's matching programmes** such as Visiting Advanced Joint Research (VAJRA) and the Global Initiative for Academic Network (GIAN). Reciprocity and two-way mobility are emphasised in the Partnership, also referring to **Erasmus+** opportunities. Moreover, **EURAXESS India** promotes EU-India cooperation by providing a one stop shop for the mobility of researchers. The Roadmap 2025 highlights the importance of strengthening cooperation in the MSCA and increasing Indian participation in the ERC to foster excellence in science and advance frontier research.

2.5.1. Marie Skłodowska-Curie Actions

The Marie Skłodowska-Curie Actions (MSCA) fund excellent R&I activities and equip researchers at all stages of their career with new knowledge and skills, through mobility across borders and exposure to different sectors and disciplines. The MSCA help build Europe's capacity for R&I by investing in the long-term careers of excellent researchers.

The MSCA also fund the development of excellent doctoral and postdoctoral training programmes and collaborative research projects worldwide. By doing so, they achieve a structuring impact on higher education institutions, research centres and non-academic organisations.

The different MSCA include Doctoral Networks (DN), Postdoctoral Fellowships (PF), Staff Exchange (SE) and Co-funding programmes (COFUND). Cooperation with India was further strengthened when on 27 August 2024, a new scheme was announced to top up selected Marie Skłodowska-Curie Actions (MSCA) Staff Exchanges projects, enabling Indian entities to carry out joint research projects with European partners and second their staff in Europe.[15]

From 2021 to 2024, a total of 417 Indian researchers and 17 entities participated in MSCA programmes (see Table 1). As illustrated in Figure 4, the primary exchange programmes were Doctoral Networks and Postdoctoral Fellowships, which accounted for nearly 90% of the participants (372 researchers). In contrast, less than 3% (11 researchers) participated through the Staff Exchange programme. It has to be noted that the figures for doctoral networks and staff exchanges are only preliminary as many selected projects had not yet started their recruitments and/or secondments at the time of writing this report.

Notably, the **participation of female researchers** from India participating in MSCA Staff Exchanges increased from 33% under Horizon 2020 to 37% under Horizon Europe, with the Doctoral Networks programme demonstrating the most progress toward gender equality.[16]

[14] EU (2021). EU-India Connectivity Partnership

[15] European Commission (2024). Boost for Indian-EU research partnerships under new co-funding scheme within MSCA Staff Exchanges

[16] European Comission (n. d.). H2020 - Marie Skłodowska-Curie Actions (MSCA) Indian Country Fact Sheet

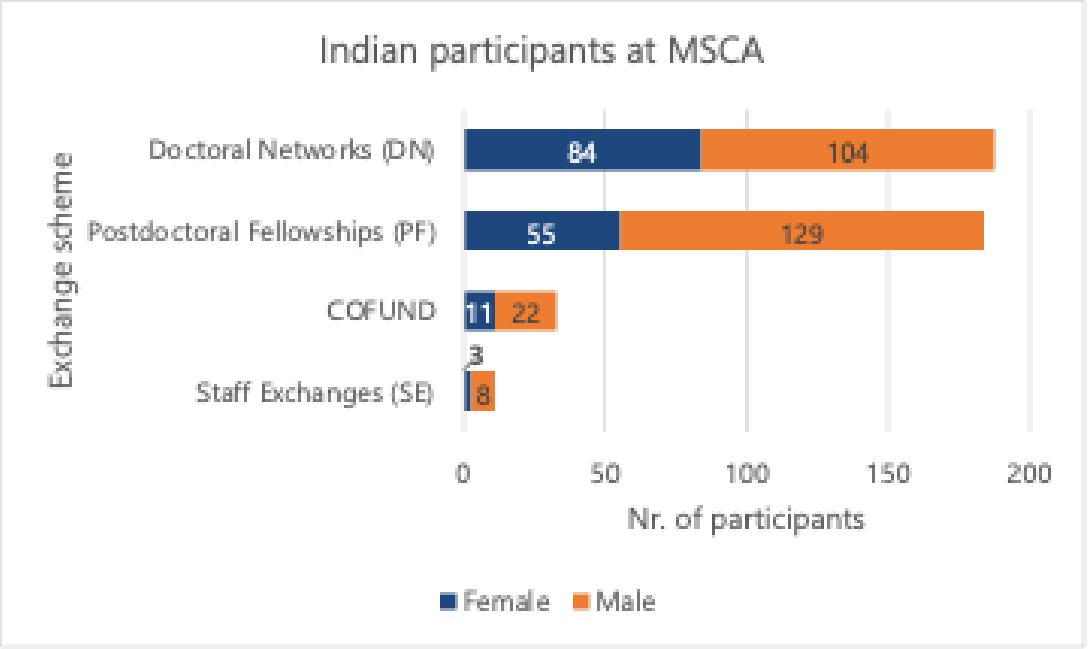


Figure 4. MSCA Participants under Horizon Europe (2021-2024)

Figure 5 provides a breakdown of the number of researchers by country of destination, with **Germany emerging as the leading host country,** with 84 Indian researchers. France and the Netherlands follow, hosting 70 and 62 researchers, respectively. This trend is similar to patterns observed under Horizon 2020, where Germany was also the top destination, followed by the UK and France. While Horizon 2020 evidenced a limited number of European researchers going to India, no data is currently available on this direction of mobility under Horizon Europe. However, anecdotal evidence suggests that there has been no significant improvement in two-way mobility.

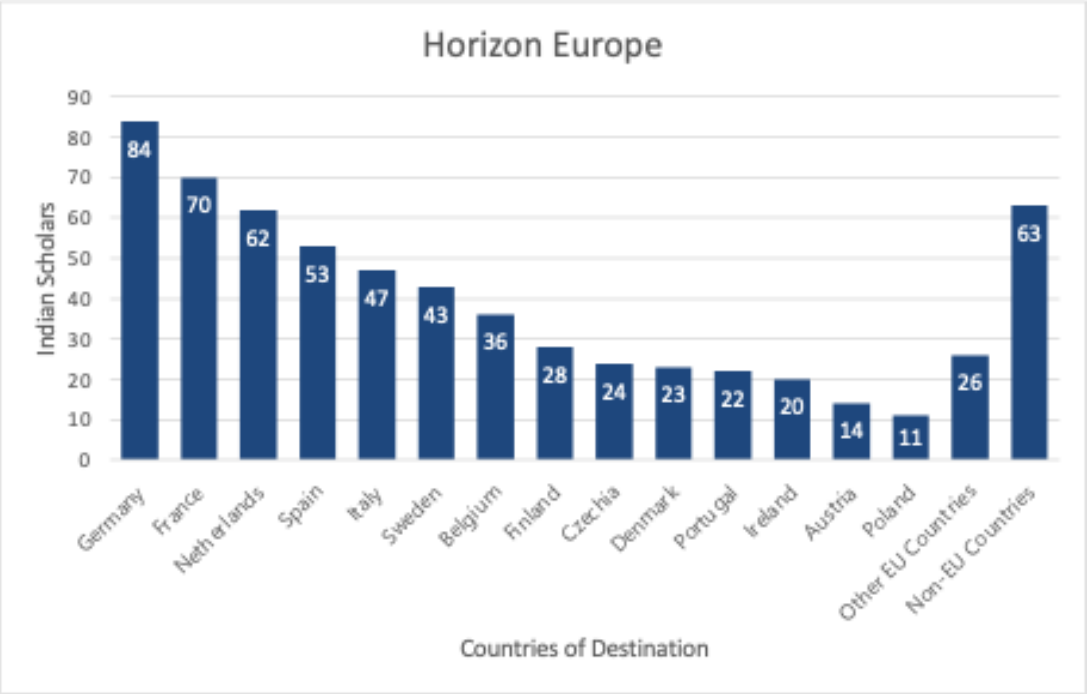


Figure 5. Countries of Destination of Indian Scholars in MSCA under Horizon Europe[17]

2.5.2. European Research Council

The **ERC** is an independent funding body established by the European Union in 2007 under Horizon 2020 and continued under Horizon Europe. It supports high-quality research across Europe through competitive funding, specifically encouraging groundbreaking scientific research. Its main grant schemes include Starting Grants for early-career researchers, Consolidator Grants for mid-career researchers, Advanced Grants for established researchers, and Synergy Grants for small groups tackling ambitious challenges. Researchers who have already received a main grant can further explore the innovation potential of their ideas by applying for an ERC Proof of Concept Grant.

Statistics on Indian nationals in ERC (2018-2023) reveal the following patterns:

- 36 ERC main grants (23 Starting Grants, 12 Consolidator Grants, one Synergy Grant)
- 10 ERC Proof of Concept Grants
- Evaluation of the first 2,000 ERC grants under Horizon 2020: 18% of the team members from non-Associated countries, with Indian nationals representing the largest share at 18%

In addition to the main grants, Indian nationals who hold active grants from the Science and Engineering Research Board (SERB) (since 2017) and the Indian Council of Social Science Research (ICSSR) (since 2020) have the opportunity to collaborate with ERC grantees through Implementing Arrangements. These collaborations can take the form of either a single long-term visit or multiple short-term visits, with funding provided by national non-EU agencies, such as science ministries. Since 2017, India ranks second in terms of interest expressed by ERC principal investigators in hosting SERB and ICSSR grantees. However, despite this high level of interest, only five SERB grantees have participated in the programme to date.

[17] When a researcher participated in an MSCA across multiple countries, each of those countries was included in the count.

3. Priority areas and main cooperation initiatives

In this section, we provide **deep dives into the S&T cooperation** between the EU and India in **priority areas** including energy, health, water, ICT and Innovation.

3.1. Energy

EU-India collaboration is key to advancing clean energy R&I. In 2016, the EU and India formed the **EU-India Clean Energy and Climate Partnership** (CECP). The EU supports this through Mission Innovation (MI), a global initiative launched at the climate summit COP 21 in Paris in 2015, involving 24 countries and the European Commission, aimed at accelerating clean and affordable energy innovation. India co-leads three and contributes to two MI Innovation Challenges focused on Smart Grids, Off-grid Access to Electricity, and Sustainable Biofuels. EU-India cooperation on energy is guided through the following **frameworks and platforms**:

- The **EU-India Roadmap 2025** emphasises collaboration on R&I, specifically targeting green and clean energy. The cooperation aims to address global challenges and support India's sustainable modernisation, with a focus on advancing clean energy transitions necessary for achieving carbon neutrality.
- Through the **TTC Working Group 2 "Green and clean energy technologies"** the EU and India are enhancing R&I to introduce sustainable technologies, thus contributing to the aim for climate neutrality by 2050 for the EU and 2070 for India. Their joint efforts in this framework focus on wastewater management, battery recycling for electric vehicles, and developing standards, while also supporting startups and skills development.[18]

Between 2019 and 2024, cooperation on energy has led to Indian participation in **7 Horizon 2020 and Horizon Europe projects**. These projects, with a total funding close to €50m, focused on the development of sustainable energy solutions, supporting climate politics, developing decarbonisation pathways, and the usage of smart grids in local energy systems (e.g. RE-EMPORWERED, see Table 2).

The collaboration between India and the European Union through initiatives like the CETPartnership and ERA-Net Smart Energy Systems has facilitated numerous projects geared towards enhancing sustainable energy solutions. Around 20 projects funded involving Indian entities through these platforms, with significant investment totalling several million euros (data not available). These projects encompass a range of innovative areas, such as renewable energy technologies, smart grids, and energy efficiency measures, providing Indian researchers and organizations with invaluable resources and expertise. Moreover, this synergy not only strengthens the bilateral ties between India and the EU but also plays a crucial role in addressing global energy challenges, aligning with both regions' sustainability goals.

Collaboration between the EU and India also includes a **partnership to advance nuclear technology**. The European Atomic Energy Community (Euratom) and India have established two key agreements. The first, signed in 2010, focuses on fusion energy research, facilitating joint projects and information exchange. The second, signed in 2020, promotes cooperation on the peaceful use of nuclear energy, including nuclear safety and waste management. Additionally, India is a member of the ITER project, a major international initiative aimed at demonstrating the feasibility of fusion power as a sustainable energy source.[19]

[18] TTC, WG2 (May 2023). Terms of Reference Working (WG) 2 on Green and Clean energy technologies.

[19] BRIDGE (2021). Cooperation between Horizon 2020 Projects in the Fields of Smart Grids, Energy Storage, Islands, and Digitalisation

RE-EMPOWERED
The "RE-EMPOWERED" project aimed to develop tools for the design and upgrading of Microgrid/Energy Island applications to high Renewable Energy Systems (RES). The project involved partners from the EU and India, with pilot sites in Denmark, Greece, and two locations in India: Keonjhar and Ghoramara. With a funding of €5m, the project had three main objectives: to increase energy efficiency and reliability, foster sustainable community development, and facilitate knowledge exchange and replicability between the EU and India. The Keonjhar pilot site, operated by IIT Bhubaneswar, features a total of 77 kWp of solar PV installations. Through integrating solar PV, biomass, storage, e-mobility, and water purification into a 40 kWp microgrid system, it is expected to provide basic electricity needs, promote livelihood activities, raise population awareness regarding renewable energies, and reduce rural-to-urban migration.

Table 2. Countries of Destination of Indian Scholars in MSCA under Horizon Europe[20] Deep dive – EU-India R&I Project on Energy (Smart Grid)

3.2. Health

India plays a crucial role in global health, particularly in health diplomacy, to achieve health-related Sustainable Development Goals (SDGs). During its 2023 G20 presidency[21], India prioritised health, with a focus on digital innovations and universal health coverage. As a leading vaccine supplier to developing countries and the world's third-largest pharmaceutical producer by volume, India's contributions in this sector are significant. However, both India and the EU face challenges in the biopharmaceutical sector, particularly due to their reliance on China for pharmaceutical ingredients. To address these challenges, it is essential for both parties to implement new policies and facilitate R&I in this sector.[22] The EU-India partnership in health has developed over more than two decades and is guided by the following **frameworks and platforms**:

- The **EU's Global Health Strategy**, which emphasises partnerships with non-EU countries.
- The **EU-India Roadmap 2025**, prioritising antimicrobial resistance, health security, and pandemic preparedness. Within the Roadmap's R&I section (Point 74), there is a strong emphasis on enhancing partnerships in health-related areas, particularly in neuroscience and brain research. This focus has facilitated multilateral initiatives, including the **Coalition for Epidemic Preparedness Innovations (CEPI)**.

The Agreement further supports this collaboration through joint and co-funded calls. Under Horizon 2020, the focus was placed on addressing critical health challenges, including tuberculosis, influenza vaccines, HIV/AIDS, malaria, and neglected tropical diseases. Efforts were also made to strengthen healthcare systems and promote universal health coverage. This collaboration resulted in three joint call projects focussing on Next Generation Influenza Vaccine Innovation (see Table 3 below), and three co-funded projects. The three joint call projects, with a total cost of €50m and over €21m from the Indian budget, aim to overcome existing scientific challenges and close research gaps in vaccine development. Two of the co-funded projects focus on tuberculosis by developing portable diagnostic technology and computer modelling for treatment, while the other concentrated on comprehensive analysis of COVID-19-related data to enhance pandemic preparedness. The DBT co-funded these initiatives, contributing a total of €3.4m to support Indian

[20] When a researcher participated in an MSCA across multiple countries, each of those countries was included in the count.

[21] Group of 20 leading industrial and emerging economies

[22] EU (2023). EU-India cooperation on health

research entities. Building on these achievements, projects in Horizon Europe continue to align with the priorities outlined in the Roadmap 2025, leading to four calls earmarked for co-funding by India. However, proposals with Indian participation were selected only in the area of pandemic preparedness and response, but not in calls related to brain disorder.

Influenza impacts millions globally, causing 500,000 deaths annually, with currently available vaccines offering only an effectiveness of about 40%. To address this, the European Commission and the DBT launched a five-year (2021-2025) collaborative programme to develop next-generation influenza vaccines.

Project	Short description
ENDFLU *Evaluation of ratioNally Designed Influenza vaccines*	ENDFLU aims to develop three next-generation influenza vaccines by leveraging European and Indian expertise. It focuses on creating constructs targeting both humoral and cell-mediated immunity. The most promising candidates will undergo phase I clinical trials in India and the EU to assess safety, immunogenicity, and optimal dosing, with further studies planned, including a controlled human infection model.
INCENTIVE *Indo-European Consortium for Next Generation Influenza Vaccine Innovation*	INCENTIVE tackles the global health and economic challenges of influenza infections. Its strategic objectives include generating crucial insights into why vulnerable individuals often respond poorly to influenza vaccines and advancing the development of two next-generation universal vaccines. This effort aims to enhance global preparedness for future pandemics.
INDIGO *Effective and Affordable Flu Vaccines for the World*	INDIGO aims to develop two innovative influenza vaccines. One combines a low-dose flu vaccine with a novel adjuvant, targeting proof-of-concept in early trials. The other uses a more immunogenic recombinant viral hemagglutinin with a potent adjuvant and needle-free delivery, aiming to significantly reduce non-responders and deliver next-generation vaccines.

Table 3. Deep dive – EU-India R&I Projects on Health (Vaccine)

3.3. Water

EU-India cooperation on water is vital for enhancing water security and sustainability, especially in India, a country that has only 4 % of the global water resources at its disposal while accounting for 18% of the global population. Cooperation on water takes place within **several framework and platforms**:

- As part of the India-EU Strategic Partnership, the **India-EU Water Partnership** (IEWP) was established in 2016 with the goal of promoting political dialogue between the EU and India. It aims to foster coherent and effective collaboration between the EU and India on water management issues by bringing together stakeholders from politics, industry and research. Within this framework, the **EU-India Water Forum** serves as key platform to facilitate dialogue, cooperation and exchange of knowledge in the field of water.

Furthermore, the **EU-India Joint Working Group on Water** was formed as a collaborative body to oversee and guide the implementation of joint water-related initiatives.[23]

- The **EU-India Strategic Partnership Roadmap to 2025** also highlights a joint interest in the field of environment and water (Point 44). It underlines the goal of fostering cooperation on water related matters through the Joint Working Group on Water and the EU-India Water Forum in the framework of the IEWP (see above).
- The first **Ministerial meeting of the TTC** in 2023 stressed the importance of EU-India cooperation in the field of wastewater management.[24] The **work plan of WG 2** (green and clean energy technologies) of 2023 specifies one action with three activities on wastewater management: Activity 1 takes up the outcomes from the existing joint projects on water (see below), exploring R&I opportunities and developing technologies for the recycling and reuse of wastewater; Activity 2 highlights plastic litter by sharing results of relevant EU projects as well as good praxis examples on monitoring and assessment of plastic waste in oceans and exploring R&I opportunities; and Activity 3 focuses on waste-to-energy, including sharing best practices, technologies and business models for waste to green hydrogen.[25]
- In 2024, an **EU-India WaterTech Event** was held bringing together stakeholders from politics, industry and research to discuss global water challenges and present innovative technologies, including those developed in joint projects (see below).

The most important R&I collaboration projects between India and the EU in the field of water were initiated through a **joint call for R&I on water** in 2017. In this context, **seven joint projects** were launched in 2018 receiving in total €40m through Horizon 2020 by the EU and by the Department of Science and Technology as well as the Department of Biotechnology of the Indian Government. The projects focus on drinking water purification, wastewater management and monitoring and control systems. Together, they developed more than 68 technologies to various TRL levels. The prioritisation of circular economy aspects in the Roadmap 2025 is less evident in the joint projects funded. One exception is PAVITR addressing reusing water for irrigation purposes. The following table presents the seven joint projects on water.[26]

Project	Short description
INDIA-H2O *Bio-mimetic and phyto-technologies designed for low-cost purification and recycling of water*	Energy-efficient and scalable solution for gaining drinking water from saline groundwater based on nature-inspired technology and powered by renewable energy
Lotus *Low-cost innovative technology for water quality monitoring and water resources management for urban and rural water systems in India*	Real-time monitoring of water supplies based on carbon nanotube technology to prevent incidences of waterborne disease

[23] EU (2020). EU-India Strategic Partnership: A Roadmap to 2025; EU (2022). EU-India Strategic Partnership. Factsheet.

[24] EU-India Trade and Technology Council (May 2023). First Ministerial meeting – Key Outcomes.

[25] TTC, WG2 (May 2023). Terms of Reference Working (WG) 2 on Green and Clean energy technologies.

[26] EU (2023). CORDIS Results Pack on water innovation in India – A thematic collection of innovative EU-funded research results. Sustainable solutions for water management.

PANI Water *Photo-irradiation and adsorption based novel innovations for water-treatment*	New technologies for making waste water safe for use in agriculture
PAVITR *Potential and validation of sustainable natural & advanced technologies for water and waste water treatment*	Cost-efficient waste water and water treatment systems
Pavitra Ganga *Unlocking waste water treatment, water re-use and resource recovery opportunities for urban and peri-urban areas in India*	Advanced technologies to treat and recycle water, contributing to the protection of local water sources from pollutants
Saraswati 2.0 *Identifying best available technologies for decentralised waste water treatment and resource recovery for India*	Different decentralised waste water treatment technologies to provide clean water to local communities
SPRING *Strategic planning for water resources and implementation of novel biotechnical treatment solutions and good practices*	Cost-efficient waste water treatment technology to clean water sources in India from pollutants

Table 4. Deep dive – EU-India R&I Projects on Water

A scoping paper on R&I cooperation between the EU and India from a European perspective in 2024 **recommends** (a) exploring replicability aspects and scalability potential; (b) examining the circular economy aspect of water; (c) further exploring future R&D opportunities and (d) strengthening capacity building and training.[27]

3.4. ICT

The EU and India are both pursuing significant digital transformation initiatives. The EU's Digital Decade, guided by the 2030 Digital Compass, aims to build a digitally empowered society across Europe. Meanwhile, India, a global leader in **Digital Public Infrastructure**, with its Digital India programme seeks to digitally empower the nation and make government services digitally accessible to all citizens. Both the EU and India have benefited from their cooperation on ICT. Since 2011, they have collaborated on ICT in the **India-EU Joint ICT Working Group**.[28] This partnership, built on regular dialogues, has focused on cybersecurity, ICT standardisation, internet governance, R&I, and 5G infrastructure. At the 2020 EU-India Summit, both parties committed to implementing the Roadmap 2025 and a **human-centric digitalisation**. The cooperation is guided through the following **frameworks and platforms**:

[27] EU (2024). Waste water Treatment and Circular Economy. Scoping paper to explore potential areas of EU-India technical R&I cooperation (European perspective).

[28] EU (2023). EU-India cooperation on digitalization. Factsheet.

- The **Roadmap 2025** has one explicit section on ICT (Points 58 to 63). This section emphasises the sector's importance by promoting common standards for digital transformation and protection of personal data, fostering cooperation on new technologies (e.g. Quantum Computing and AI), and strengthening collaboration between EU and Indian industries and startups in innovation and technology deployment.
- The **TTC Working group 1 "Strategic Technologies, Digital Governance and Digital Connectivity"** subsumes previous joint tasks and working groups in the sector. The mutual areas of interest identified build upon previous agreements. In the first **Ministerial meeting of the TTC** in 2023 three key initiatives were defined: Coordination of the Global Partnership on AI (GPAI), coordination of policies for the semi-conductor sector, and the development of Digital Public Infrastructure (see Table 5).[29]

Under the co-funded call earmarked by Indian agencies in the **Horizon Europe Work Programme 2023-2024, "**Efficient trustworthy AI - making the best of data (AI, Data and Robotics Partnership) (RIA)", **2 proposals with Indian participation were selected. Co-funded by the DST, the Indian Institute of Technology is participating in the "Reliable AI and Data Optimisation Project" (RAIDO). RAIDO is a powerful framework designed to develop trustworthy and green artificial intelligence, with a total budget of €8m.**

Initiative	Short description
Artificial Intelligence Coordination	India and the EU are collaborating through the GPAI to bridge AI theory and practice. India is focused on maximising the benefits of AI while preventing its misuse, aligning with both the EU's AI Act and India's AI initiatives under the India-EU Roadmap to 2025.
Semiconductor Policy Coordination	India and the EU signed a memorandum of understanding (MoU) in November 2023 aimed at diversifying semiconductor supply chains and reducing reliance on a single source. The MoU supports collaboration, with the EU's Chips Act and India's incentives (approx. ₹84b, equivalent to €900m) to become semiconductor manufacturing hubs with robust and sustainable supply chains.
Digital Public Infrastructure	Recognising the digital initiatives of both partners and the importance of India's Digital Public Infrastructure, their collaboration in the TTC focuses on improving interoperability and advancing sustainable development, especially in developing nations.

Table 5. Deep dive – EU-India R&I Initiatives on ICT

3.5. Innovation

As recommended in the previous review, two innovation pilot initiatives were launched between the EU and India, mostly benefitting startups and entrepreneurs.

The first project, the **Europe-India Innovation Partnership**, was launched in the context of the European Commission's Service Facility for International Cooperation. It ran from 2018-2021 and brought together a mix of stakeholders to leverage innovation for sustainable socioeconomic development: startups and entrepreneurs, startup incubators and accelerators, investors,

[29] EU (2024). EU-India Trade and Technology Council. Factsheet.

corporate partners, and EU and Indian policymakers. The initiative promoted the exchange of ideas and mutual understanding of each side's innovation ecosystems through a series of networking events. Through the incubators and accelerators connected, more than 400 innovators and other stakeholders were reached in total. As a result, a 'food waste and loss' thematic challenge was launched, several European Innovation Zones were set up as soft-landing hubs in Bengaluru and Hyderabad in India, and several corporate partners were scouted for European and Indian startups to collaborate on innovative solutions to shared problems of sustainable development. The impact of the initiative was somewhat limited due to the onset of the Covid-19 pandemic, which made in-person interaction impossible.

The Europe-India Innovation Partnership was followed up by the **Europe-India Innovation Center**, a Horizon 2020 project that ran from 2021 until 2023. The Innocenter focussed on helping European startups and small and medium enterprises (SMEs) access the Indian market by helping them find suitable local partners for sustainable partnerships. The project managed to support 26 European companies create around 100 partnerships with Indian startups and other stakeholders. Through market validation, the European companies were able to access the Indian market, leading to sales, increased production, research and other forms of collaboration. The main sectors were health, energy, electric vehicles and cloud services, agro/agritech, education and FinTech. The selected companies contributed to the Indian economy by hiring local talent, building revenue streams and forging partnerships with Indian private and public sector partners.

The project concluded with a review, according to which the following lessons can be learnt: Focus on targeting a smaller number of companies specifically, do not neglect the potential of startups based in Eastern Europe, accelerate startup support, use industry and pilot projects to raise awareness for market potential, have the EU level work closely with the Member State level in India, focus on mature startups and do not neglect the importance of technical standards for facilitating market entry. In the future, it would be useful to also help Indian startups access European markets. Product-market fit and the creation and execution of an India-specific market strategy are important for European companies. Sector-specific approaches are required, e.g. for green hydrogen, electric mobility, renewable energy, water treatment, and waste processing.[30]

Together, the connections established through **these two projects between European and Indian entrepreneurs and innovators form a basis for launching more specific innovation cooperation initiatives** targeting priority areas of the TTC, such as hydrogen or circular economy, in the future. In particular, the thematic challenge approach piloted in the Europe-India Innovation Partnership, but also the targeted, sector-specific startup support piloted by the EU-India Innocenter could be revived for this purpose.

[30] EU (2024). EU India INNOCENTER Policy Recommendations Report (updated version)

4. Trade and Technology Council

The EU and India have strengthened their relationship as strategic partners through the establishment of a **TTC** by agreement between the President of the European Commission Ursula von der Leyen and the Indian Prime Minister Narendra Modi on 25 April 2022 in New Delhi. As the second such bilateral forum for the EU, following the EU-US TTC established in 2021, the EU-India TTC aims to increase bilateral cooperation, foster trade and investment, and ensure technological and industrial leadership of the partners.[31]

The TTC currently comprises **three working groups** (WG):

1. **Strategic technologies, digital governance and digital connectivity,** led by the Ministry for Electronics and IT (India) and DG CNECT (EU): current topics include digital connectivity, artificial intelligence, 5G/6G, high performance and quantum computing, semiconductors, cloud systems, cybersecurity, digital skills and digital platforms.
2. **Green and clean energy technologies,** led by the Office of the Principal Scientific Adviser to the Government of India (India) and DG RTD (EU): current work focuses on green technologies, including investment and standards, with a focus on R&I. Areas to be explored include clean energy, circular economy, waste management, plastics and marine litter. It will also promote cooperation between incubators, SMEs and startups in the EU and India.
3. **Trade, investment and resilient value chains,** led by the Ministry of Commerce (India) and DG TRADE (EU): current topics include supply chain resilience and access to key components, energy and raw materials, removal of trade barriers and global trade challenges. In addition, cooperation in multilateral bodies will be worked on to promote international standards and cooperation in tackling global challenges.

As part of the TTC, one **Ministerial meeting** in May 2023 and one **Stocktaking Call** in November 2023 took place. As part of Working Group 2 (WG2), one **International Workshop** was held in October 2023. In the Ministerial meeting in May 2023, both parties presented the main areas of cooperation under the three WGs of the TTC and (for WG2) agreed on a Work Plan In the workshop in October 2023, a Roadmap for WG2 was discussed for five main areas of cooperation: waste to green hydrogen, marine plastic litter and waste water, e-mobility and battery circularity, charging infrastructure interoperability, and standards for R&I. The Roadmap for WG2 was then validated in the Stocktaking Call in November 2023. These mutual agreements and efforts set the basis for substantial work plans for scientific and technological cooperation over the next few years.

Based on the conclusions of these meetings and (for WG2) the agreed Roadmap, the following **activities relevant to R&I** have been carried out up until late 2024:

Under **WG1,** the EU and India strengthened cooperation across several key areas relevant to R&I, including semiconductors, high-performance computing (HPC), digital skills and infrastructure, AI, IT standardisation and 6G technology. Under **WG2**, a **Work Plan** was adopted at the first Ministerial meeting in May 2023 focusing on three Actions: (1) Tackling wastewater management, including waste to hydrogen and marine plastic pollution, (2) E-mobility, battery recycling and charging infrastructure, (3) Standards. DG RTD of the European Commission and the Office of the Principal Scientifc Advisor (PSA) to the Government of India (GoI), co-chairs of the WG, met in July 2023 and agreed to prioritise the workplan and to conduct the above-mentioned International Workshop of WG2 in October 2023 focusing on waste to green hydrogen, marine plastic litter and waste management, e-mobility and circulatory aspects of batteries and session as well as interoperability of charging infrastructure and standards.

[31] EU (2024). EU-India Trade and Technology Council. Factsheet

Further, **short-, medium and longer-term plans for Working Group 2** were developed. The **short-term plan** included (1) a training module at the Joint Research Centre (JRC) E-Mobility Lab in Ispra, Italy (2) a matchmaking event for Indian and EU startups, (3) an e-learning training network connection and (4) exploring Automotive Research Association of India (ARAI)-JRC collaboration. All the planned activities have been implemented to date: In January 2024, a successful two-week mutual exercise took place. Two Indian scientists/engineers of ARAI attended a training module on e-vehicle batteries, energy storage and smart mobility at the JRC Ispra Lab on European Interoperability Centre for Electric Vehicles, Smart Grid and Smart Homes. It focussed on sharing achievements and research progress, particularly with regards to standards for charging infrastructures, allowing for a knowledge exchange on the convergence of charging capacity and the potential for cooperation.

A Water Tech Event took place in January 2024 in Mumbai, bringing together the joint research projects on water and enabling an exchange on the emerging technologies developed under the projects and their possible application in the Indian context.

In addition, a matchmaking event with a focus on recycling of batteries took place virtually in June 2024. Three EU and three Indian startups were selected to conduct market exploration trips in India and in the EU, which took place in September/October 2024. The startups met investors, companies, government agencies and clusters. They are now exploring with partners on possible activities such as joint R&D, setting up of joint recycling plants, or investment opportunities. This model is intended to be replicated for other areas under WG2, such as plastic litter.

An online workshop on "*Assessment and Monitoring Tools for Marine Plastic Litter"* was organised in November 2024, bringing together experts and officials from India and the EU to address plastic pollution and discuss innovative monitoring solutions. The workshop focused on exploring new digital tools for monitoring plastic waste and understanding India's unique challenges in combating plastic litter.

The JRC has initiated discussions with the Indian counterpart ARAI on technical and regulatory aspects of electric mobility. As part of this engagement, a joint hybrid workshop on "EV Charging Technologies Standardisation Strategies" took place on 24-25 February 2025 in ARAI´s premises in Pune (India) and online. This workshop brought together Indian experts, regulatory and testing authorities, and industry representatives to enhance the EU-India dialogue and foster industry participation in the standardisation process for electric vehicle charging technologies in India.

In 2024, a €1.4m EU project was launched to support the implementation of WG2 through joint activities such as workshops, exchanges of experiences and studies.

The **medium- to long-term plan for WG2** includes joint science and technology cooperation in marine plastic litter, waste-to-renewable hydrogen, battery recycling for electric vehicles, and wastewater treatment technologies. These projects are expected to foster EU-India collaboration among researchers, innovators, and technologists to develop sustainable and cost-effective solutions, Further cooperation will focus on harmonising EV charging infrastructure standards, pre-normative research for harmonised testing solutions, and promoting knowledge exchange on science-based standardisation

Activities under WG3 are not directly relevant to R&I and are thus not covered by this review.

A **2nd TTC Ministerial meeting** taking place in New Delhi in late February 2025 will review progress under the three Working Groups and confirm the expected deliverables for the next period.

5. Conclusions and recommendations for future cooperation

The **main conclusions** that can be drawn from this review are outlined below:

Strategic alignment

1. Overall, the review shows a **positive trend for S&T cooperation between India and Europe**. The S&T Agreement, in place since 2002, provides a solid foundation, reinforced now with the creation of the TTC, through its WGs and Ministerial meetings. This allows a regular refining of the focus of cooperation and joint initiatives.
2. Overall, **the strategic areas of cooperation** are in line with the Roadmap 2025 and remain the same (focus on Environment/Climate, Energy and Health) in both programmes. In Horizon Europe the strategic areas seem to be more diversified, as there are more calls/projects in e.g. ICT and biodiversity (in line with recommendation of previous review). WG 2 of the TTC contributes significantly to advancing collaborative efforts in green and clean energy technologies, waste management, and the circular economy. Yet, there are areas which are explicitly mentioned in the roadmap and in the last review (e.g. cultural heritage and brains disorder) that have not been addressed so far in projects under the co-funded calls earmarked by Indian agencies. This could indicate a **discrepancy between political ambitions and the actual involvement of Indian partners** in relevant calls. It is therefore recommended that the EU indicate India as a target country in co-funded, earmarked calls and specify available co-funding for Indian participants in upcoming Horizon Europe Work Programmes.
3. The absence of one single structured and consolidated **Indian R&I programme** that would be equivalent to the EU's Horizon Europe framework programme may make it harder to agree on thematic priorities. However, collaboration between the EU and India could be further enhanced by exploring opportunities through EU-funded thematic Joint Programming Initiatives (JPIs) and ERA-Nets. By leveraging these platforms, India can engage with a broader network of EU member states, reducing fragmentation and fostering synergy in research and innovation efforts. This collaborative approach can lead to more significant impacts on joint projects, aligning regional agendas with global sustainability goals. Such strategic partnerships would not only promote knowledge exchange but also enhance resource mobilization, creating a more cohesive framework for tackling urgent energy challenges and advancing shared objectives between India and Europe in the context of clean energy and climate action.
4. Aligning future cooperation with the Roadmap and the commitments taken by the JSTCC and building on the results of the work of the EU-India TTC could be one avenue to address the aforementioned issues (Points 2 and 3). Moreover, the **midterm review of Horizon Europe** could provide an opportunity to reinforce **joint research cooperation** with the Global South and with India in particular.

Participation, visibility and funding modalities

5. There was a **continuously high number of collaborative actions** within the Horizon programmes, yet the share of participation of Indian partners decreased over time. It is recommended to adopt clear rules on budget allocation for joint calls to prevent any discrepancies between EU and Indian co-funding that may lead to reductions in budget during implementation.
6. The **Flagship calls** on water, vaccines and smart grids stand out through their joint efforts leveraging large Indian involvement regarding impact, participation and funding.
7. Yet, a **decrease in overall project funding** and in particular in **Indian contribution and participation** can be observed. This decrease might be related to the **shift from joint**

calls under Horizon Europe to the CFM, which does not require the inclusion of Indian partners. Although the 2016-2020 review of EU-India S&T cooperation recommended simplifying funding rules to encourage participation[32], the transition to the CFM has not noticeably improved Indian involvement. The EU's Horizon Europe Impact Assessment Plan33 highlights that increasing participation from Third Countries, such as India, depends heavily on good communication and tailored rules in the calls for proposals. The European Climate, Infrastructure and Environment Executive Agency (CINEA) also outlined the importance of tailored rules to avoid frictions and guarantee efficient collaboration in a contribution paper on energy joint calls between the EU and India. Overall, the fact that Indian participation in CFM calls remains optional may be a key factor contributing to this negative trend.

8. For Horizon Europe projects, **EC and Government of India funding rules and processes** could be more closely aligned. Joint calls have the potential to enhance Indian participation. Evaluations could be conducted during project implementation to allow for adaptations to project implementation.

Mobility of researchers and innovators and capacity-building

9. There is a large number of Indian participation and grantees in **MSCA** and **ERC**. The high number of Indian PhD students and Post-Docs participating in MSCA helps achieve the objective of promoting scientific excellence through EU-India cooperation (in line with the recommendation of the last review). The high number of Indian ERC grantees shows the drive for frontier research. Among EU countries, Germany remains a top destination for Indian researchers (MSCA) and there seems to be a trend towards a higher number of female researchers involved. However, there is no evidence of an improvement with regards to the two-way mobility that had been recommended in the previous review. It is thus important to extend the scope of student exchange programmes and skills development to support EU-to-India mobility.

10. Several innovation initiatives have successfully piloted activities in support of European and Indian entrepreneurs and startups collaborating on solutions to shared challenges of sustainable socioeconomic development. While the impact of these projects was somewhat limited due to the Covid19-pandemic, the connections established through them provide a fertile base for further initiatives to be launched in the future, possibly targeting priority areas of the TTC WGs.

[32] EU (2021). Agreement for scientific and technological cooperation between the EU and India Review of cooperation between May 2016 – May 2020

[33] European Commission (2018) A New Horizon for Europe: Impact Assessment of the 9th EU Framework Programme for Research and Innovation

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This report evaluates the EU-India Science and Technology (S&T) Agreement, set to be renewed for 2025–2030. EU-India S&T cooperation has evolved through Horizon 2020 and Horizon Europe, with Indian participation in 88 projects. Key areas of collaboration include energy, health, water, information and communication technology (ICT), and innovation. Joint research initiatives have focused on smart grids, vaccine development, wastewater treatment, and artificial intelligence. The Trade and Technology Council (TTC) facilitates cooperation on green technologies, digital governance, and trade. There was a continuously high number of collaborative actions within the Horizon programmes, yet the share of participation of Indian partners decreased over time due to shifts in funding models, from joint calls to a co-funding mechanism. Human capital development and mobility has been strengthened through Marie Skłodowska-Curie Actions (MSCA) and European Research Council (ERC) grants, but mobility from Europe to India could be higher. The report recommends clearer budget allocations, alignment with the TTC and building on strategic partnerships to reinforce cooperation.

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